Shangze (Rudy) Dai

Ph.D. Candidate in Applied Economics

MCCA 1172, 1692 McCarty Dr.
Gainesville, FL 32603
1-352-575-5109

shangzedai@ufl.edu
Personal Website
Google Scholar

Research Interests

Economics of Innovation and Technological Change; Economics of AI; Industrial Organization and Production Economics; Environmental and Resource Economics.

Education

Ph.D. in Applied Economics, University of Florida — 2023 – 2027 (expected)
Research area: Economics of New Technology; GPA: 3.95/4.00

M.S. in Applied Mathematics, Johns Hopkins University — 2023 – 2026
Field: Applied Analysis and Stochastic Processes; GPA: 3.84/4.00

M.Phil. in Economics, Wuhan University — 2020 – 2023
Field: Regional Economics; GPA: 3.82/4.00

Certificate in International Studies, Hopkins-Nanjing Center — 2022 – 2023
Field: International Economics; GPA: 3.91/4.00

B.S. in Economics, Tongji University — 2016 – 2020
Field: Mathematical Finance; GPA: 4.65/5.00

Current Research

The Economic Value of Precision Agriculture under Drought Risk: Evidence from U.S. Corn Counties

- Examines whether precision agriculture mitigates the effects of drought on corn yields and crop insurance losses using county-level data from major U.S. corn-producing states.
- Combines county-year panel models, entropy balancing, staggered difference-in-differences, machine learning, and a dynamic stochastic welfare framework.
- Finds that precision agriculture reduces drought-related production and insurance losses and generates estimated benefits of approximately \$5–\$20 per acre per year.

Dissertation

Theme: Geopolitical Risk, Technology Catch-up, Energy Transition, and Food Security

- Food Security Concerns under Geopolitical Risk: Land, Trade, and Ecology.
- Geopolitical Risk and Renewable Energy Transition.
- U.S.–China Technological Decoupling and Indigenous Innovation in China.

Peer-Reviewed Publications

- **Dai, S.*** & Ji, J. (2026). Unintended carbon cost of automation technology in transforming economies: The role of capital dependence and structure change. *Ecological Economics.*
- **Dai, S.***, Ji, J., & Yu, H. (2026). Technology Import and Biased Technological Progress in Transforming Economies: Evidence From China. *Review of Development Economics.*

- Yang, B., Fan, F.*, **Dai, S.***, et al. (2026). Industrial Robot, Resource Misallocation, and Firm Innovation Performance. *Journal of Regional Science.*
- **Dai, S.*** (2025). Understanding automation's impact on ecological footprint. *Environmental and Resource Economics.*
- Li, G., Zhang, N., **Dai, S.**, et al. (2025). Trade liberalization, factor network association, and energy poverty. *Energy Economics.*
- Cao, Y. & **Dai, S.*** (2025). Understanding and evaluating new quality productive forces in China: based on machine learning. *Applied Economics Letters.*
- **Dai, S.**, Dai, Y., Hu, J., & Yu, H. (2025). Energy-saving technological change, regional economy growth and endowment structure: empirical evidence from Chinese cities. *Technology Analysis & Strategic Management*, 1–14.
- Fan, F., Yang, B., Cao, Y., & **Dai, S.*** (2024). The spatial effects of innovation agglomeration on air quality in China: from the perspective of industrial development. *International Journal of Sustainable Development & World Ecology*, 1–18.
- Yu, H., **Dai, S.**, & Ke, H. (2024). Industrial collaborative agglomeration and green economic efficiency—Based on the intermediary effect of technical change. *Growth and Change*, 55(2), e12727.
- **Dai, S.**, Yu, H., Aiya, F., & Yang, B. (2024). Green bonds and sustainable development: theoretical model and empirical evidence from Europe. *International Journal of Sustainable Development & World Ecology*, 1–16.
- **Dai, S.**, Dai, Y., & Yu, H. (2024). The effect of gender gap in labor market participation on carbon emission efficiency: state level empirical evidence from the US. *Energy and Environment*, 1–27.
- Fan, F., Zhang, K., **Dai, S.**, & Wang, X. (2023). Decoupling analysis and rebound effect between China's urban innovation capability and resource consumption. *Technology Analysis & Strategic Management*, 35(4), 478–492.
- Fan, F., **Dai, S.**, Yang, B., & Ke, H. (2023). Urban density, directed technological change, and carbon intensity: An empirical study based on Chinese cities. *Technology in Society*, 72, 102151.
- Ke, H., **Dai, S.**, & Fan, F. (2022). Does innovation efficiency inhibit the ecological footprint? An empirical study of China's provincial regions. *Technology Analysis & Strategic Management*, 34(12), 1369–1383.
- Ke, H., Yang, B., & **Dai, S.*** (2022). Does intensive land use contribute to energy efficiency?—evidence based on a spatial durbin model. *International Journal of Environmental Research and Public Health*, 19(9), 5130.
- Ke, H., **Dai, S.**, & Yu, H. (2022). Effect of green innovation efficiency on ecological footprint in 283 Chinese Cities from 2008 to 2018. *Environment, Development and Sustainability*, 24(2), 2841–2860.
- Fan, F., **Dai, S.**, Zhang, K., & Ke, H. (2021). Innovation agglomeration and urban hierarchy: Evidence from Chinese cities. *Applied Economics*, 53(54), 6300–6318.
- Ke, H., **Dai, S.**, & Yu, H. (2021). Spatial effect of innovation efficiency on ecological footprint: City-level empirical evidence from China. *Environmental Technology & Innovation*, 22, 101536.

** Corresponding author.*

Working Papers

- **Dai, S.**, Ji, J., & Huang, K.* Mitigating the Heat: The Role of AI Technology in Shielding Corporate Environmental Performance from Extreme Temperatures. *R&R at Business Strategy and the Environment.*
- Yu, H. & **Dai, S.*** Rural E-commerce and land-augmenting technological progress. *R&R at Land Use Policy.*

- **Dai, S.**, Ji, J., & Huang, K.* Signaling Corporate Resilience under Extreme Weather Shocks: Strategy, Visibility, and Market Responses. *R&R at Business Strategy and the Environment.*
- **Dai, S.**, Lu, C.*, & Li, G.* Pollution-Induced (Mis-)Reallocation in Research: Evidence from U.S. Water Quality Shocks. *R&R at The B.E. Journal of Economic Analysis & Policy.*

Teaching Experience

PhD Math Camp Instructor, Summer 2025
University of Florida (FRE); Evaluation: 4.9/5
Course materials

AEB 3103 Principles of Food and Resource Economics Teaching Assistant, Fall 2025
University of Florida; Instructor: James Ji

AEB 4138 Advanced Agribusiness Management Teaching Assistant, Spring 2025
University of Florida; Instructor: Jacklyn Kropp

ECO 7115 Microeconomic Theory Teaching Assistant, Fall 2024
PhD Core Course, University of Florida; Instructor: Fatma Gunay

AEB 2014 Economic Issues, Food and You Teaching Assistant, Fall 2023, Spring 2024
University of Florida; Instructor: Jennifer Clark

Presentations

- **2026**: *Berkeley/Sloan Summer School in EEE*, Berkeley (2026.08); *AAEA Annual Conference*, Kansas (2026.07); *CES NA Conference*, Atlanta (2026.03); *SAEA 2026 Conference*, Louisville (2026.2);
- **2025**: *SEA 2025 Conference*, Tampa (2025.11); *CES CN Conference*, Guangzhou (2025.07); *CAERE Annual Conference*, Shanghai (2025.05); *Seminar on Climate Change Economics*, Online (2025.01)
- **2024**: *Seminar on Environmental Economics*, Online (2024.11)
- **2023**: *China Regional Economics Annual Forum*, Nanjing (2023.07)
- **2022**: *Jiangsu Young Economists Forum*, Nanjing (2022.11); *Future Economist Forum*, Beijing (2022.11)
- **2021**: *China Innovation Economic Forum*, Wuhan (2021.06); *Seminar on Urban and Rural Development*, Nanjing (2021.05)
- **2020**: *China Spatial Economics Annual Conference*, Guangzhou (2020.12)

Awards and Honors

- *Sloan Foundation Travel Scholarship*, ARE, UC Berkeley 2026
- *AEM/GSS Case Study Competition Second Place*, AAEA 2026
- *J. R. Greenman Memorial Scholarship*, CALS, University of Florida 2025
- *Harold B. Clark Award*, FRE, University of Florida 2024
- *Grinter Fellowship*, FRE, University of Florida 2023
- *Amerisolar Scholarship* (Top 3), Hopkins Nanjing Center 2023
- *National Scholarship for Graduate Students* (Top 1), Wuhan University 2022

Academic Services

- Ad hoc reviewer for more than 20 journals, including: *Nature Cities*, *Energy Economics*, *Journal of Business Research*, *Technovation*, *Cities*, *International Review of Financial Analysis*, *Technology in Society*, *Growth and Change*, and *Transportation Research Part E.*
- Academic reputation survey respondent for *Times Higher Education* and *U.S. News & World Report.*

Professional Service, Outreach, and Experience

- *Leadership*: *Academic Vice President, AE-Graduate Student Organization* (2025–2026).
- *Outreach Publication*: Coauthored “Diverging Paths: How Latin America Is Reorienting Exports amid Geopolitical Tensions,” *Choices Magazine* (2026).
- *Volunteer*: *UF Water Institute Symposium* (2024), *World Laureates Forum II* (2019).
- *Internship*: *CITIC Construction Investment Co., Ltd.* (2019).

Skills

- *Languages*: English, Chinese, French (Basic).
- *Programming*: R, Python, Stata, MATLAB, SQL, LaTeX.

Professional Memberships

- American Economic Association (AEA); Agricultural & Applied Economics Association (AAEA); Chinese Economist Society (CES)

References

Dr. Zhengfei Guan (Chair)
Associate Professor
Food and Resource Economics Department
University of Florida
E-mail: guanz@ufl.edu

Dr. Fan Fan (Committee Member)
Assistant Professor
Food and Resource Economics Department
University of Florida
E-mail: ffan@ufl.edu

Dr. Gunnar Heins (Committee Member)
Associate Professor
Department of Economics
University of Florida
E-mail: gheins@ufl.edu

Dr. James Ji (Committee Member)
Assistant Professor
Department of Food, Agricultural and Resource Economics
University of Guelph
E-mail: james.ji@uoguelph.ca

Dr. Kurt Stein (External Referee)
Associate Professor of Practice
Academy of Data Science
Virginia Tech
E-mail: kstein@vt.edu

Dr. Patrick Ward (Committee Member)
Associate Professor
Food and Resource Economics Department
University of Florida
E-mail: wardp@ufl.edu